

Pre-Activated 3D CNN and Feature Pyramid Network for Traffic Accident Detection

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Abstract—In this paper, we present a novel traffic accident detection method using a spatio-temporal three-dimensional (3D) convolutional neural network. The proposed method consists of pre-activation ResNet and feature pyramid network (FPN) structure. To reduce computational load with preserving the detection accuracy, we propose interval input method. Experimental results show that the proposed network outperforms existing methods in the sense of both precision and recall measures.

Index Terms—C3D, traffic accident, interval input

I. INTRODUCTION

According to the recent survey by the Ministry of Public Administration and Security of South Korea, CCTV newly installed for crime prevention and traffic information collection is increasing every year. All CCTVs are monitored by control agents and agents take action when an abnormal situation occurs. There are various situations in abnormal situation, but when a traffic accident occurs, the damage is generally large and immediate response is needed. If the control agents can monitor only the CCTVs suspected of an traffic accident situation, the efficiency of the CCTV control system can be increased. Therefore, many studies are progressing for the development of a traffic accident detection system applying various image processing techniques and deep learning to maximize the efficiency of the control system using CCTV by automatically detecting the occurrence of a traffic accident and preventing the second accident through quick response. Existing studies have proposed dividing normal-flow and abnormal-flow based on optical flow, which is one of the methods of calculating the motion in video. Then, whether the abnormal-flow region is a vehicle or not is classified based on deep learning algorithms [1] [2]. These existing methods are scale-variant and have a low accuracy rate.

In this paper, we present a method to detect traffic accidents in CCTV video with high detection rate compared to existing method using modified pre-activation structure, spatio-temporal 3D kernel and FPN structure in residual network [3].

II. PROPOSED METHOD

Our method enables parallel processing through CNN architecture to improve the many computing times of existing methods and increases accuracy by using the pre-activation ResNet and FPN structure to overcome low accuracy and scale-variant problem of existing method.

A. 3D-Convolution neural network

Generally, 2-dimensional CNN has superior performance in image, but it does not use time information, so it is difficult to process video data efficiently. To overcome this limitation, the 3D Convolution Neural Network(C3D) uses spatio-temporal 3D kernels with convolution and pooling to use both space and time features of video data [4] [5].

In this paper, 3D residual network (3D-ResNet) is constructed by using spatio-temporal kernel based on the structure of 2D residual neural network. Our interval input method consists of 16 frames in total, skipping by 3 frames.

Figure 1 shows the difference between the conventional input method and the proposed interval input method. By using this interval method, the information between consecutive frames in the video is very small, thus minimizing the loss of information and reducing the number of input parameters, thereby helping to computation time while maintaining accuracy.

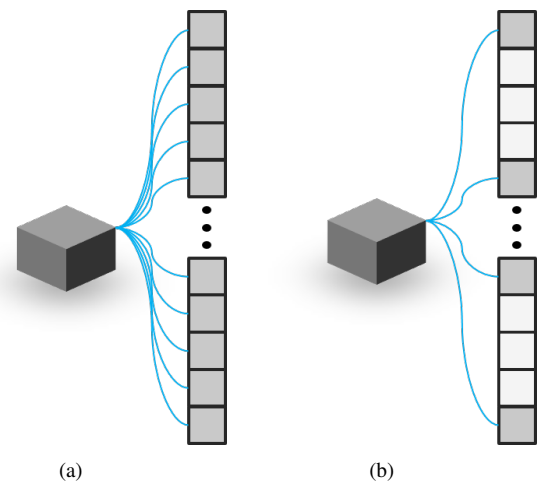


Fig. 1. Comparison of input methods : (a) conventional input method, (b) proposed interval input method

B. Pre-activation residual network

Our proposed method uses a modified pre-activation(PA) structure. The basic pre-activation structure simplify the calculation of the backward path by moving the ReLU following each skip connection process to the inside of the residual block

to improve the performance of the back propagation process of the network [6].

The method we propose uses a modified pre-activation structure that additionally removes the first ReLU and adds BN before the skip connection, as shown in Figure 2. This modified pre-activation structure helps the network learn better than the basic pre-activation structure [7].

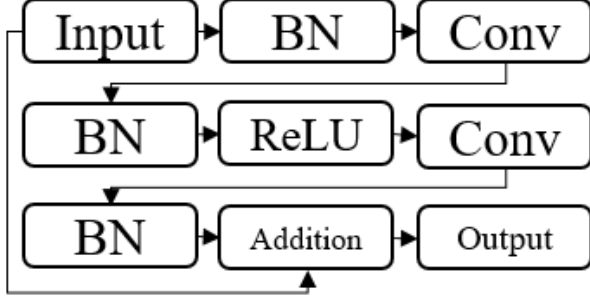


Fig. 2. Modified pre-activation residual block structure

C. Feature-pyramid network structure

To solve the scale-variant problem of the existing method, the FPN structure has been applied in this paper. Unlike a typical network using only the output of the last layer, FPN structure uses features extracted from multiple layers of the network, as shown in Figure 3 [8]. In this paper feature map P_{3-5} extracted from C_{3-5} of the ResNet Structure consisting of the five large layers were applied to the FPN structure. This structure helps the network become scale-invariant by supplementing the lost information as the network goes through convolutions. The feature maps extracted from the network are used by combining the information of the different feature maps after growing the size through the up-sampling process.

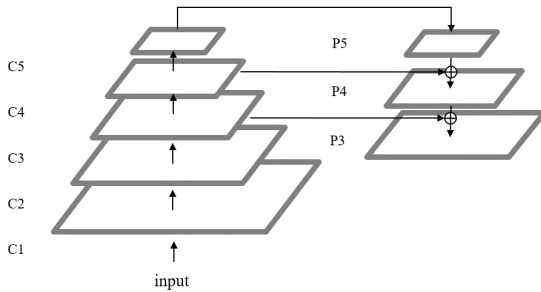


Fig. 3. Feature-Pyramid Network structure

III. EXPERIMENTAL RESULTS

There are no public datasets related to traffic accidents, and datasets of existing studies have collected videos through cooperation with specific companies or via YouTube.

We collected video clips taken on CCTV for experiments via YouTube. The collected videos consist of 25 fps on average.

We constructed the training data with 114 traffic accident videos and 152 non-accident videos, and set the epoch to 200 based on ResNet 34. The test data for experiment consisted of 44 accident videos and 72 non-accident videos.

The proposed method took 2.63 seconds to process a video with a 7 second length of 600*480 resolution of 25 fps and took 9.61 seconds without an interval input.

Table I shows the results of classifying a total of 116 videos by the proposed method and shows the performance of the 3D-ResNet baseline and ablation study. Experimental results show higher precision than the existing methods based on optical flow. The result of comparing the proposed method with the existing method is shown in Table II.

The method proposed in this paper is expected to increase the efficiency of the CCTV control system because it has a high detection rate and a scale-invariant advantage over the conventional method which uses a rather classical existing method. Furthermore, proposed methods showed inaccurate tendency in the occlusion situation. Therefore, it is necessary to study how to utilize depth information of images.

TABLE I
EXPERIMENTAL RESULTS

	Experimental Method		
	3D ResNet	with PA	with PA+FPN
Accident	66/84	68/84	71/84
Normal	107/143	113/72	117/72

TABLE II
COMPARISON RESULTS

	Experimental Method		
	Proposed Method	B. Maaloul	V. E. M. Arecada
Precision	0.82	0.68	0.75
Recall	0.84	0.91	0.80

IV. CONCLUSION

In this paper, we proposed a method to detect traffic accidents in CCTV video using modified pre-activation ResNet, spatio-temporal 3D kernel and FPN structure in residual network. Since the proposed structure is based on CNN, it is possible to improve the processing speed by enabling parallel processing. The interval input helps to reduce the computational load, and the modified pre-activation and the FPN are helping to improve accuracy compared to existing methods. This method can be applied to CCTV control system to help increase the efficiency of the control system by increasing the number of CCTVs monitored by each control agent.

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